Shellfish (crustaceans and bivalve molluscs): HMTc Standard

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CITE AS

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for shellfish, crustaceans (shrimp, crab, lobster) and bivalve molluscs (mussels, clams, oysters, scallops), not canned, stated as consumed. Bivalves concentrate cadmium, and the cadmium ceiling reflects the strictest applicable government maximum; every limit is the strictest maximum level any credible government sets for the product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	500	government maximum level
Cadmium (Cd)	Tier 1	500	government maximum level
Total arsenic (tAs)	Tier 2	1,000	government maximum level
Inorganic arsenic (iAs)	Tier 1	100	government maximum level

Mercury, total (tHg)	Tier 1	500	government maximum level
Methylmercury (MeHg)	Tier 1	500	human-grade read-across (Mexico)
Nickel (Ni)	Tier 2	2,000	human-grade read-across (European Union)
Tin (Sn)	Tier 2	200,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	100,000	human-grade read-across (China)
Total chromium (Cr)	Tier 2	1,000	government maximum level
Hexavalent chromium (Cr-VI)	Tier 2	10	human-grade read-across (US-CA)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	government maximum level
Cadmium (Cd)	500	government maximum level
Total arsenic (tAs)	1,000	government maximum level
Inorganic arsenic (iAs)	100	government maximum level
Mercury, total (tHg)	500	government maximum level
Methylmercury (MeHg)	500	Mexico — human-food analog
Nickel (Ni)	2,000	European Union — human-food analog
Tin (Sn)	200,000	European Union — human-food analog
Aluminium (Al)	100,000	China — human-food analog
Total chromium (Cr)	1,000	government maximum level
Hexavalent chromium (Cr-VI)	10	US-CA — human-food analog

4 | Cite this standard

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References

[1] China — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni, Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Mexico — human-food maximum levels
Binds MeHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] US-CA — human-food maximum levels
Binds Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/shellfish. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

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